

Ângulo de Antonio Vandr .

Sejam $P(a, b)$, $Q(c, d)$, o eixo $\overrightarrow{PQ}$, e uma fun  o $f : I \rightarrow \mathbb{R}$. O  ngulo θ de um ponto de f com o eixo $\overrightarrow{PQ}$   tal que

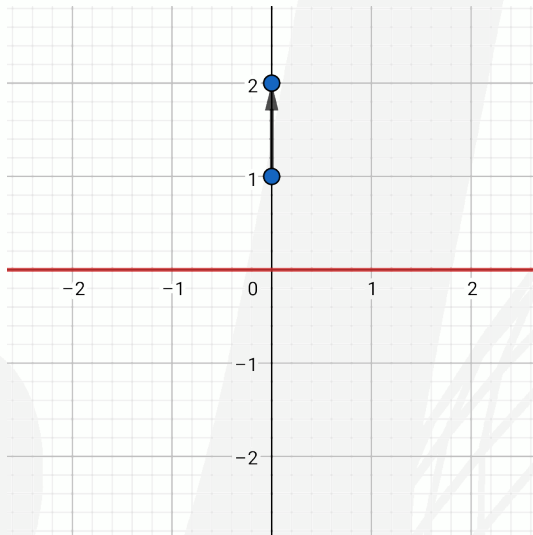
$$\cos \theta = \frac{(c-a)(x-a) + (d-b)[f(x)-b]}{\sqrt{[(c-a)^2 + (d-b)^2]\{(x-a)^2 + [f(x)-b]^2\}}}.$$

Chamando tal  ngulo de  ngulo de Antonio Vandr ,

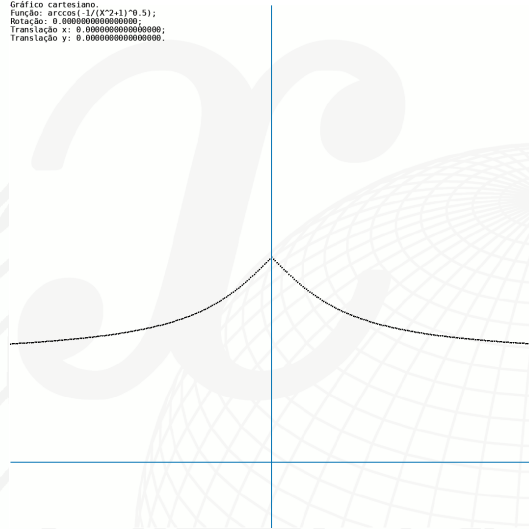
$$\alpha_{\mathcal{A}_{f(x)}}^{[(a,b),(c,d)]} = \arccos \frac{(c-a)(x-a) + (d-b)[f(x)-b]}{\sqrt{[(c-a)^2 + (d-b)^2]\{(x-a)^2 + [f(x)-b]^2\}}}.$$

Exemplo: $f(x) = 0$, $P(0, 1)$, $Q(0, 2)$:

$$\alpha_{\mathcal{A}_0}^{[(0,1),(0,2)]} = \arccos \frac{-1}{\sqrt{x^2 + 1}}.$$



Gr fico cartesiano.
Fun  o: $\arccos(-1/(x^2+1))^{0.5}$;
Bordado: 0.0000000000000000;
Transla  o x: 0.0000000000000000;
Transla  o y: 0.0000000000000000.



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